

Working Procedure: Target Prioritization in Lung Adenocarcinoma (LUAD)

Step 1: Disease Selection

Lung Adenocarcinoma (LUAD) was selected as the disease of interest due to its high prevalence among non-small cell lung cancers and the availability of large-scale genomic and clinical datasets.

Step 2: Data Source Identification

To perform target prioritization, publicly available and curated biological databases were identified based on their relevance to disease–target association analysis.

The following databases were selected:

- **Open Targets Platform** – primary source of disease–target association data
 - **DisGeNET** – validation of disease–gene relationships
 - **GWAS Catalog** – assessment of inherited genetic associations
 - **UniProt** – functional annotation and druggability information
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Step 3: Dataset Download

From the Open Targets Platform:

- Disease selected: *Lung Adenocarcinoma (EFO_0000571)*
- Dataset downloaded: **Disease–Target Association TSV file**
- File name used: openTarget_dataset.tsv

This dataset contains gene-level evidence scores derived from genetics, cancer driver databases, somatic mutations, clinical biomarkers, pathways, and literature support.

Step 4: Data Loading

The downloaded TSV file was imported into Python using the pandas library.

Each row represented a candidate gene, and each column represented a specific biological evidence type.

Step 5: Feature Selection

Only biologically relevant evidence columns were retained for analysis, including:

- globalScore
- cancerGeneCensus
- intogen
- evaSomatic
- cancerBiomarkers
- chembl
- reactome
- europepmc

Irrelevant or redundant fields were excluded.

Step 6: Data Preprocessing

- Missing values were replaced with zero to indicate absence of evidence.
- Genes with weak disease association were removed using a threshold:

globalScore ≥ 0.2

This filtering step ensured that only LUAD-relevant genes were retained for downstream analysis.

Step 7: Evidence Weight Assignment

Biologically informed weights were assigned to each evidence feature.

Higher weights were given to:

- Genetic association
- Cancer driver evidence

Moderate weights were assigned to:

- Clinical and somatic mutation data

Lower weight was assigned to:

- Literature frequency

This weighting strategy reduced bias toward well-studied but biologically non-essential genes.

Step 8: Priority Score Calculation

A composite priority score was calculated for each gene using a weighted sum of all selected evidence features.

This score represents the overall biological and translational relevance of the target in LUAD.

Step 9: Target Ranking

Genes were ranked in descending order based on their final priority scores.

Top-ranked genes were identified as high-confidence therapeutic targets.

Step 10: Result Interpretation

The prioritized targets were interpreted based on:

- Known oncogenic driver roles
- Involvement in resistance mechanisms
- Therapeutic tractability

Key prioritized targets included:

EGFR, KRAS, MET, BRAF, STK11, and TP53.

Final Outcome

This workflow enabled systematic identification of biologically validated and clinically relevant targets in lung adenocarcinoma, supporting rational decision-making for early-stage drug discovery.